Bayesian Co- and Postseismic Slip Model of the 2023 M_w 7.8 and 7.7 Türkiye–Syria Earthquake Doublet from Geodetic Data

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Leonid Pereiaslov[*1], Théa Ragon[1,2], Yuan-Kai Liu[1,3], Mary Grace Bato[4], Marin Govorcin[4], Eric Fielding[4], and Mark Simons[1]

Abstract

On 6 February 2023, an earthquake doublet occurred in the East Anatolian fault zone (EAFZ), producing significant surface displacements and causing extensive damage in Türkiye and Syria. The EAFZ is known to be spatially complex, with multiple segments, fault stepovers, and fault-bend structures. How such fault complexities modulate the co- and postseismic distribution of fault slip remains an important question. We use Sentinel-1 and Advanced Land Observation Satellite-2 Synthetic Aperture Radar-derived pixel offset measurements and Global Navigation Satellite Systems (GNSS) static offsets for the two events to constrain the distribution of coseismic slip along both strands of the EAFZ. We constrain the postseismic slip distribution using Sentinel-1 interferograms and GNSS time series. We adopt a Bayesian framework to explore the solution space, relying solely on physically justifiable prior information without employing artificial regularization. We account for both observational errors and epistemic uncertainties due to imperfect knowledge of the regional elastic structure and orientation of the fault plane. We find that elastic afterslip mostly surrounds the areas of coseismic rupture. However, the geodetic data sets that span the postseismic period cannot be explained solely by elastic afterslip. We also confirm the potential seismic gap on the boundary between the Erkenek and Pütürge faults, across which neither coseismic nor postseismic slip occurs in the periods considered here.

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Supplemental Material

Introduction

On 6 February 2023, two earthquakes devastated the borderland between Türkiye and Syria. The M_w 7.8 left-lateral mainshock, nucleating on a previously unknown fault, progressed over a ~300 km long zone from the Mediterranean coast of Türkiye to the north, toward the city of Çelikhan (Melgar *et al.*, 2023). This earthquake was followed approximately 9 hr later by an M_w 7.7 magnitude sinistral aftershock, 90 km to the north–northeast of the mainshock (Melgar *et al.*, 2023).

These two earthquakes ruptured two strands of the East Anatolian fault zone (EAFZ), which, along with the North Anatolian fault zone (NAFZ), serves as a boundary of the westward extrusion of the Anatolian plate (McKenzie, 1972; Dewey *et al.*, 1986). South of the city of Çelikhan, the EAFZ bifurcates into two strands, the northern strand known as the Sürgü fault, and the southern strand known as the Erkenek fault (Duman and Emre, 2013). Although NAFZ has hosted a series of destructive events during the past century (e.g., Barka and Kadinsky-Cade, 1988; Tibi *et al.*, 2001), the EAFZ has been less active in terms of strong earthquakes (Konca *et al.*, 2021). The 2020 M_w 6.8 Elazığ earthquake, which ruptured the Pütürge segment, was the largest event to occur on the EAFZ since the 1971 M_w 6.8 Bingöl earthquake (Pousse-Beltran *et al.*, 2020; Konca *et al.*, 2021; Ragon *et al.*, 2021; Fig. 1).

Large earthquakes have occurred along the EAFZ during both historical and instrumental periods. Several strong earthquakes have ruptured the southern strand, including the 521 M_w 7.5 Amanos, 1513 M_s 7.4 Pazarcık and 1893 M_s 7.1 Erkenek events (Ambraseys and Jackson, 1998; Duman and Emre, 2013; Fig. 1). Each of these earthquakes ruptured a distinct fault that is well defined geologically and characterized by structural irregularities at its boundaries (Duman and Emre,

1. Seismological Laboratory, California Institute of Technology, Pasadena, California, U.S.A., https://orcid.org/0009-0005-7508-1065 (LP); https://orcid.org/0000-0002-1276-1910 (TR); https://orcid.org/0000-0003-4032-9444 (Y-KL); 2. Now at Université Grenoble Alpes, Université Savoie Mont Blanc, CNRS, IRD, Université Gustave Eiffel, ISTerre, Gières, France; 3. Now at Department of Geophysics, Stanford University, Stanford, California, U.S.A.; 4. Jet Propulsion Laboratory, California Institute of Technology, Pasadena, California, U.S.A., https://orcid.org/0000-0002-6648-8067 (EF)

*Corresponding author: lpereias@caltech.edu

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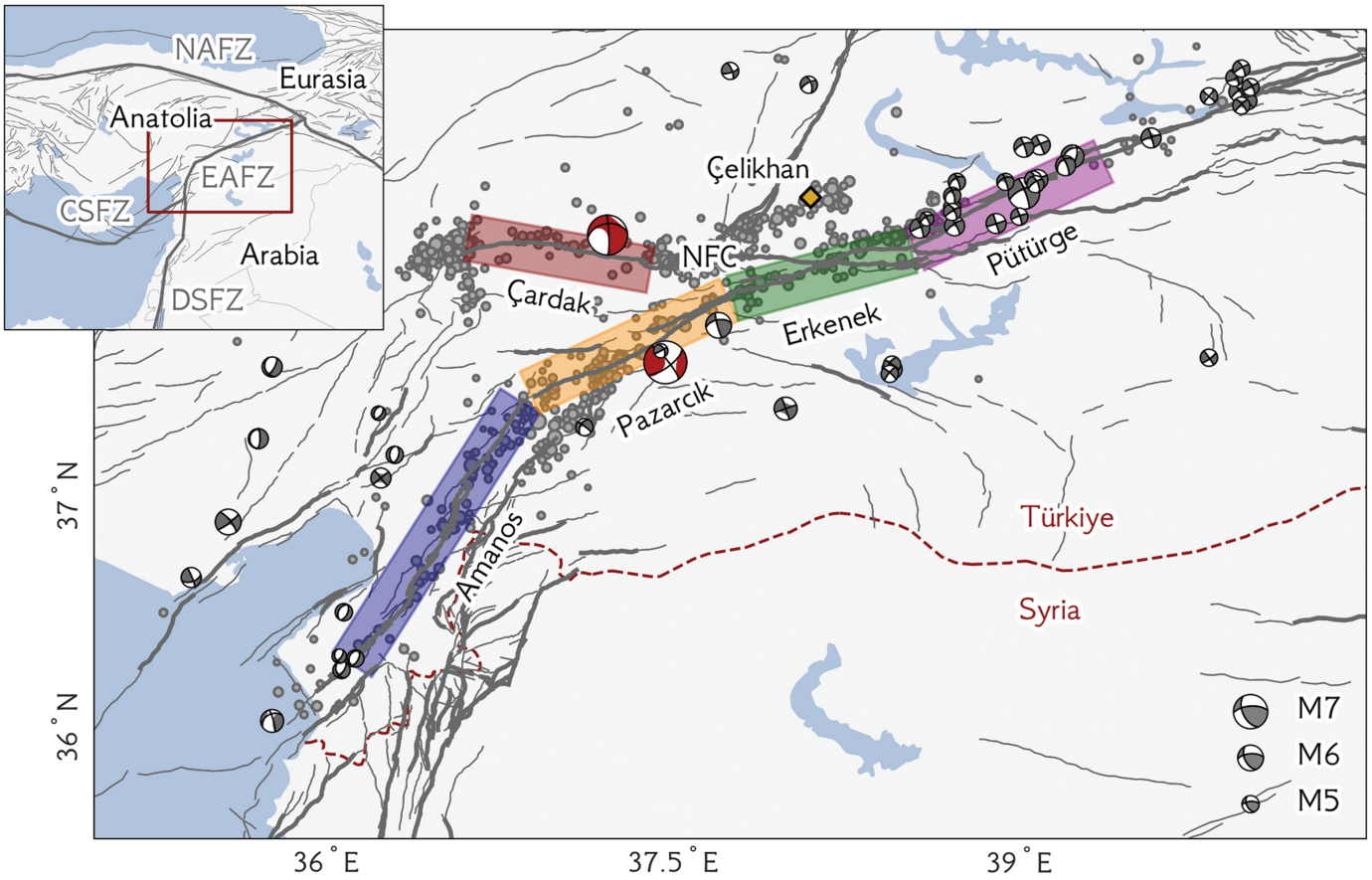


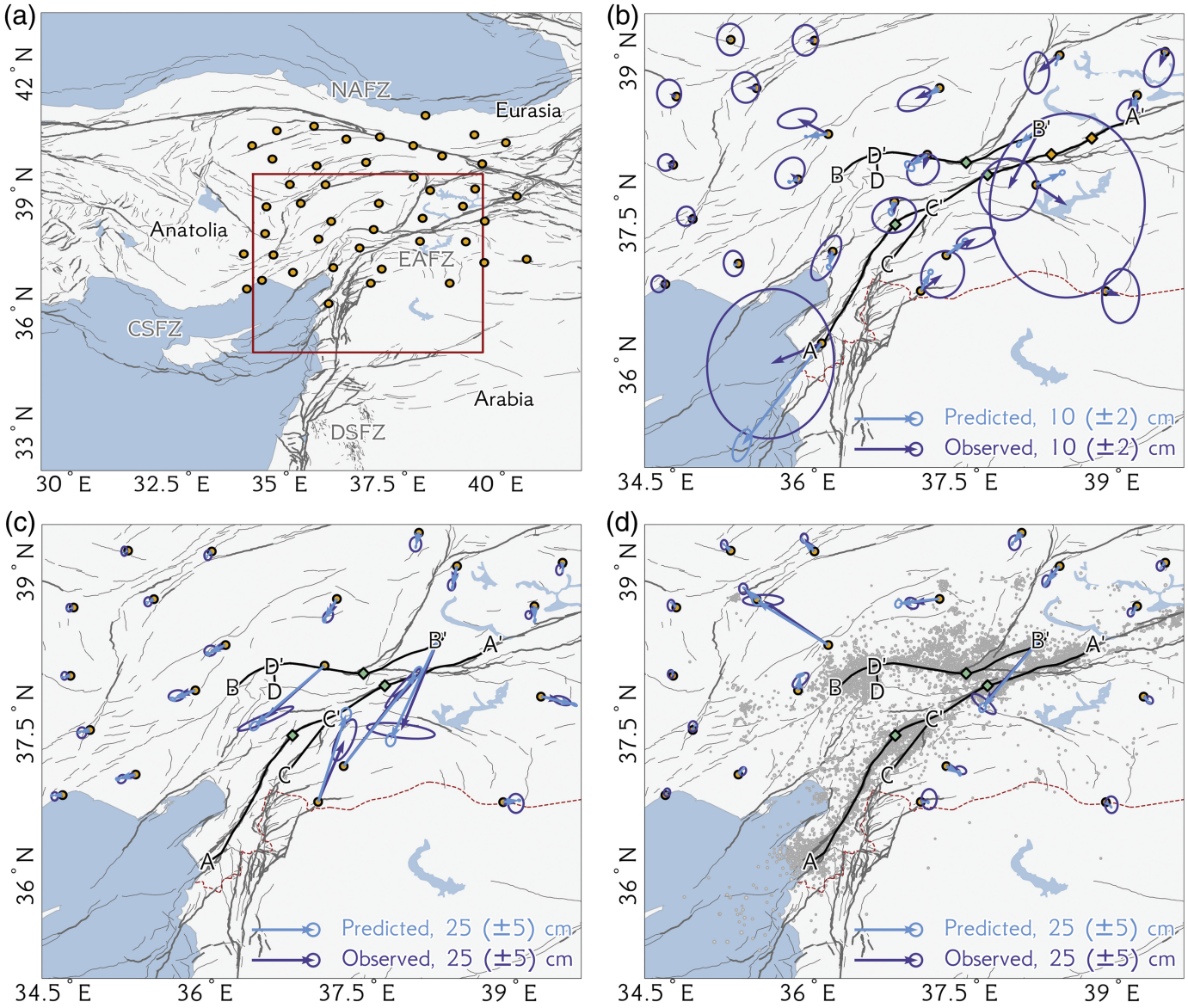
Figure 1. Tectonic map of the region. Fault segmentation, based on the historical earthquakes as well as geology, is displayed in colors (Duman and Emre, 2013; Güvercin *et al.*, 2022). Blue: 521 M_w 7.5 and possibly 1822 M_s 7.5 Amanos Fault earthquakes. Orange: 521 M_w 7.5 and 1513 M_s 7.4 Pazarcık Fault earthquakes. Green: 1893 M_s 7.1 Erkenek fault earthquake. Magenta: 2020 M_w 6.8 Pütürge fault earthquake. Red: 1544 M_s 6.8 Çardak fault earthquake. Faults are presented according to the active faults of Eurasia database (Zelenin *et al.*, 2022). NFC, Nurhak fault complexity. Moment tensor solutions are obtained using IRIS web services (Trabant *et al.*, 2012). Gray dots indicate aftershocks, which are taken from the Turkish Seismic Network Catalogue (see Data and Resources). The inset shows the broader plate-tectonic setting, with plate boundaries and major fault zones; the box marks the area of the main map.

2013). The Türkoğlu restraining bend separates the Amanos and Pazarcık faults, whereas the Gölbaşı releasing stepover separates the Pazarcık and Erkenek faults. (Duman and Emre, 2013). Those fault complexities of the southern strand are marked with green diamonds in Figure 2. A devastating earthquake with magnitude that may have exceeded 7.8 occurred in 1114 on the southern strand (Ambraseys and Jackson, 1998; Sbeinati *et al.*, 2005), which would roughly correspond to the part of the EAFZ that ruptured during the 2023 M_w 7.8 event. The 2023 M_w 7.7 earthquake occurred on the northern strand of the EAFZ, for which no reports of large earthquakes in the last 2000 yr exist, with the exception of the 1544 M_s 6.8 event (Duman and Emre, 2013). The Amanos, Pazarcık, and Erkenek faults, which ruptured during the 2023 M_w 7.8 earthquake, had been suggested to have the potential for a large earthquake before the actual event (Nalbant *et al.*, 2002; Duman and Emre, 2013; Aktug *et al.*, 2016). Based on the interseismic slip-rate partitioning between the EAFZ strands and the recurrence intervals of strong events on the Pazarcık segment, Duman and Emre (2013) proposed a recurrence interval of 800–1000 yr for M_w 7.0 or stronger earthquakes on the eastern half of the northern strand of the EAFZ. The occurrence of the 2023 M_w 7.7 event is consistent with these inferences.

Estimates of interseismic slip deficit rate found that the eastern part of the EAFZ has a significant slip deficit in the first kilometers of the upper crust (down to a depth of approximately 5 km; Cavalié and Jónsson, 2014; Bletery *et al.*, 2020). The interseismic finite-fault inversion also showed that the shallow interseismic slip deficit varies along strike and that the 2020 M_w 6.8 Elazığ earthquake likely ruptured a portion that previously was inferred to have a significant slip deficit (Bletery *et al.*, 2020).

The western part of the EAFZ, which ruptured during the 2023 M_w 7.8 event, has been significantly less studied, and detailed models of the interseismic slip deficit do not exist for that section. Walters *et al.* (2014) included data from the western part of the EAFZ and estimated that a significant slip deficit extends to a depth of approximately 15 km uniformly along the whole EAFZ, which is a rough estimate. A more recent study by Aktug *et al.* (2016) found that the depth

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extent of the slip deficit along the western segment is poorly constrained with the currently available data, but estimated it to be 25 km. However, all recent geodetic studies agree fairly well on the level of strain accumulation along this segment, suggesting a relative motion along the EAFZ of the order of 1 cm/yr (Reilinger *et al.*, 2006; Walters *et al.*, 2014; Aktug *et al.*, 2016).

Many models of the earthquake doublet have been published (Barbot *et al.*, 2023; He *et al.*, 2023; Jia *et al.*, 2023; Liu *et al.*, 2023; Mai *et al.*, 2023; Melgar *et al.*, 2023; Okuwaki *et al.*, 2023; Zhao *et al.*, 2023; Chen *et al.*, 2024; Ma *et al.*, 2024). There is general consensus that the mainshock started on the secondary fault close to the Pazarcık segment and bilaterally ruptured the Amanos and Erkenek segments while propagating across the fault bends and releasing stepovers. The aftershock ruptured the Çardak fault, bilaterally as well, extending further into the previously unknown fault across the Nurhak fault complexity (NFC), shown with a green diamond in

Figure 2. (a) Tectonic map of the region showing locations of all the Global Navigation Satellite Systems (GNSS) stations used. (b) GNSS observations as well as predicted displacements of the mean models for the joint postseismic slip, (c) mainshock, and (d) the aftershock. Full error covariance of GNSS observations is displayed with confidence ellipses at two-sigma levels. The distribution of displacements of all the models of the Bayesian ensemble is shown with similar ellipses. Green diamonds represent boundaries between different segments. These boundaries are marked with vertical lines in Figures 4 and 5. Golden diamonds show the approximate boundaries of the seismic gap mentioned in Barbot *et al.* (2023), where the eastern boundary is chosen based on the aftershock and afterslip distribution of the 2020 Pütürge earthquake (Cakir *et al.*, 2023). These boundaries are also marked with vertical lines in Figures 4 and 5. Gray dots indicate aftershocks, which are taken from a relocated catalog (Lomax, 2023).

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Figure 2. In general, it displayed a slip distribution more compact along strike, compared with the preceding mainshock. However, key questions remain unresolved, such as the depth of the rupture or the details of its distribution along the complicated ensemble of faults, due to significant variability between the models. Kinematic rupture models, based on the high-rate (HR)-Global Navigation Satellite Systems (GNSS) and strong-motion (Melgar *et al.*, 2023) and teleseismic (Okuwaki *et al.*, 2023) data sets, show significant slip (around or exceeding 5 m) reaching a depth of 20 km. Some geodetic-only (Jia *et al.*, 2023) and joint far- and near-field seismic and geodetic models (Jia *et al.*, 2023; Liu *et al.*, 2023) display similar results. In contrast, most geodetic-only (Barbot *et al.*, 2023; He *et al.*, 2023; Mai *et al.*, 2023; Zhao *et al.*, 2023; Ma *et al.*, 2024), one of the teleseismic models (Mai *et al.*, 2023) and a joint HR-GNSS, strong-motion and static GNSS offset model (Chen *et al.*, 2024) confine the slip to the shallower crust, with a peak slip above 10 km depth. Some models show a more or less continuous slip along the entire rupture length (He *et al.*, 2023; Jia *et al.*, 2023; Okuwaki *et al.*, 2023; Zhao *et al.*, 2023; Chen *et al.*, 2024), whereas other models are characterized by a patch-like pattern (Barbot *et al.*, 2023; Liu *et al.*, 2023; Mai *et al.*, 2023; Melgar *et al.*, 2023; Ma *et al.*, 2024).

A wide range of factors can contribute to the differences between the preferred solutions. The choice of data sets and their relative weights, the assumed elastic structure and fault geometry, as well as the size and shape of the subfaults are all impacting the final result of the finite-fault inversion. Moreover, conditions used to impose regularization in least-squares inversion are often nonphysical; that is, it is usually difficult to justify either the smoothness of the solution or its proximity to a particular prior model. To avoid these pitfalls, we employ an alternative approach, based on exploring the full model space while trying to account for the epistemic uncertainties (Minson *et al.*, 2013).

Ergintav *et al.* (2024) constrained the immediate postseismic slip for the first few months after the earthquake doublet. However, that study only uses GNSS observations and models the fault planes with low spatial resolution. Guo *et al.* (2025) used Interferometric Synthetic Aperture Radar (InSAR) data to constrain the coseismic slip and the afterslip during the first four months after the event. In that study, the authors present a regularized least-squares model with uncertainty estimated through a jackknife sensitivity analysis. We revisit both the co- and postseismic models. We investigate the relation of co- and postseismic slip, modeled without regularization, along the fault branches to obtain a spatial as well as a temporal image of the underlying processes within three months after the event. We also reexamine the lateral extent of the coseismic slip to understand the relationship between the along-strike limits in rupture extent and the stepover and bend structures of the EAFZ. In addition, we evaluate the extent to which afterslip may fill in a proposed seismic gap between the Erkenek and Pütürge segments (Barbot *et al.*, 2023).

Coseismic Geodetic Data

Satellite pixel offset data

We use coseismic azimuth and range offsets from four Synthetic Aperture Radar (SAR) pairs, three Sentinel-1 (A014, A116, and D021) and one Advanced Land Observation Satellite-2 (ALOS-2; D078) satellite tracks (where A and D indicate ascending and descending portions of an orbit; Table S1, available in the supplemental material to this article). Both data sets have been processed using the InSAR Scientific Computing Environment (ISCE2) software augmented with an additional module to process ALOS-2 data (Rosen *et al.*, 2012; Liang and Fielding, 2017). In general, pixel offset tracking analysis provides surface displacements with lower precision and higher noise than line-of-sight interferometric measurements (Simons *et al.*, 2002). However, strong disruption of the surface close to the fault trace, dense vegetation, and snowy conditions all contribute to the reduced correlation of the interferometric phase data. For these reasons, phase unwrapping the interferograms for this region is challenging, and thus interferometric phase data are not included in the inversion for the distribution of coseismic slip.

We downsampled the SAR pixel offset data to use them efficiently in the inversion, while retaining as many features of the original displacement field as possible (Figs. S1–S3). We use a quadtree algorithm that gives preference to subdividing regions with high curvature of the Green's function. The approach is based on an algorithm described in Wang *et al.* (2014), and similar to that originally described in Lohman and Simons (2005). We remove data points located closer than 2 km to the fault traces because our subfaults are too large to fit those data points without introducing modeling artifacts. To calculate the covariance matrix $\mathbf{C_d}$, for the SAR data, we use the residual map derived from a preliminary regularized least-squares inversion (Jolivet *et al.*, 2012; Yue *et al.*, 2017) and then fit the variogram cloud with a Gaussian variogram model (Lohman and Simons, 2005; Chiles and Delfiner, 2009; Sudhaus and Sigurjón, 2009), which, assuming second-order stationarity, is then converted through a linear relationship to the covariance (Fig. S4).

GNSS data

We use GNSS coseismic static offsets and the corresponding uncertainties to constrain our models. The displacements and their associated uncertainties are obtained from the UNR Geodetic Laboratory (Blewitt *et al.*, 2018). However, we exclude two stations from the slip inversion. EKZ1 is excluded from the aftershock inversion because it is located too close to the fault trace, and its signal may therefore be influenced by complex off-fault deformation processes that are not explicitly included in our model. CYL2 is excluded from both events because, for unknown reasons, its displacement is clearly anticorrelated with that of the surrounding stations. This leaves 36 stations for the mainshock and 35 for the aftershock. The GNSS data allow us to separate surface displacements between the two events, whereas the SAR images combine the effects of both earthquakes. We

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consider only the horizontal components of the GNSS observations, due to the low signal-to-noise ratio (SNR) of the vertical displacements.

The covariance matrix $\mathbf{C_d}$ for the GNSS data needs special consideration. We find that, although the overall slip distribution changes little, deep slip artifacts appear along the lower edge of the model when we adopt the $\mathbf{C_d}$ for the GNSS data processing method proposed by the UNR Geodetic Laboratory (Fig. S5). These artifacts might be caused by the complex motions of individual GNSS stations that cannot be properly represented by the simplified elastic slip model. As such, the standard deviations of the GNSS observations are multiplied by 3 (raising the uncertainties to the average of 8 and 11 mm for the eastern and northern components, respectively) to reconcile those observations with the uncertainties of the pixel offset data sets (which are 30 cm on average; Fig. 2). This choice of scaling allows the reduced chi-squared value for GNSS observations to approach 1. Leaving the uncertainties as they were originally provided or scaling them by 2 leads to a chi-squared value significantly higher than 1. In effect, this uncertainty upscaling is an ad hoc form of accounting for epistemic or prediction errors.

Postseismic Geodetic Data

GNSS data

For the postseismic inversion, we use the time series of local GNSS stations, with daily solutions processed by the UNR Geodetic Laboratory (Blewitt *et al.*, 2018). As with the coseismic inversion, we employ only the horizontal components. We use the DISSTANS Python package (Köhne *et al.*, 2023) to decompose the time series into secular, annual, semiannual, co- and postseismic components. The coseismic offset of the earthquake doublet on the date of 6 February 2023 is represented with a step function, and a postseismic model, composed of the two logarithms with time constants equal to 1 and 10 days, provides a good fit to the time series (Figs. S6 and S7). This choice of time constants is made for fitting purposes; they are not to be interpreted on their own. We employ such a parametric fit because it enables us to estimate the displacement at a specific date while simultaneously utilizing information from the entire time series. By jointly modeling the secular trend, seasonal signals, coseismic offsets, and postseismic deformation, the estimate at the target date is constrained by the complete temporal evolution of the station. In this sense, it can be regarded as a geophysically informed interpolation. We exclude the station FEEK from the set due to the time series not being well fit while simultaneously displaying significant displacement values, leaving 43 stations for the postseismic inversion. We consider displacements during the first three months after the earthquake in the inversion. Uncertainties, which are obtained by the least-squares solution of the DISSTANS package, are scaled for the postseismic GNSS data set by 8, bringing them to 11 and 13 mm for the eastern and northern components, respectively, and also comparable to the coseismic GNSS uncertainties. This choice is based on the same reduced chi-squared value logic discussed earlier. The model obtained using unscaled GNSS uncertainties is shown in the supplement (Fig. S8). However, we note that our models still have difficulty fitting the postseismic GNSS observations (Fig. 2). That is, to get the reduced chi-squared value for this data set to approach 1, one must increase the GNSS uncertainties higher than the InSAR uncertainties (which are equal to 6 mm on average), thereby reducing the importance of the GNSS data, although the InSAR data are spatially correlated and the GNSS data are not.

Satellite InSAR data

Postseismic InSAR time series are calculated for the three Sentinel tracks (A014, D021, and A116; Fig. S9). For the A014 track, there are 82 pairs over the time range of 9 February to 24 November 2023. For the D021 track, there were 82 pairs over the time range of 10 February to 13 November 2023. For the A116 track, 82 pairs were made over the time range of 28 February to 19 November 2023. We use the ISCE2 software to process these interferometric pairs, and then we employ small baseline InSAR time-series analysis using the MintPy package (Yunjun *et al.*, 2019). A postseismic model includes a logarithm with a time constant equal to 10 days is fitted to those time series for consistency. We omit a logarithm with a time constant equal to 1 day, since the InSAR time series have significantly lower temporal coverage immediately after the events as compared to GNSS. Postprocessing is the same as described for the coseismic satellite data sets in terms of noise estimation (Fig. S4) and downsampling (Fig. S10). We mask out the southwest part of the A014 and D021 data sets because there is a region with a short-wavelength component of the displacement field, which is not consistent with being driven by our fault model (Fig. 3). This region of deformation is probably related to unaccounted-for off-fault processes or the effect of the 20 February 2023 M_w 6.4 aftershock, which we do not consider here.

Model

Fault geometry

Funning *et al.* (2007) suggested that complex fault planes need to be represented with as much precision as possible because simplifying the fault geometry introduces a significant bias in recovering the slip distribution (Dutta *et al.*, 2021). In our case, the western and central segments of the northern branch, as well as the entire southern branch, are mapped through the combination of SAR and optical correlation datasets and the active faults of Eurasia database (Zelenin *et al.*, 2022).

The eastern segment of the northern branch is more difficult to model because the lack of surface rupture allows for multiple interpretations on whether the two branches should be connected (Melgar *et al.*, 2023; Okuwaki *et al.*, 2023) or not (Barbot *et al.*, 2023; Jia *et al.*, 2023; Liu *et al.*, 2023; Mai *et al.*, 2023). The aftershocks of the second event clearly locate along

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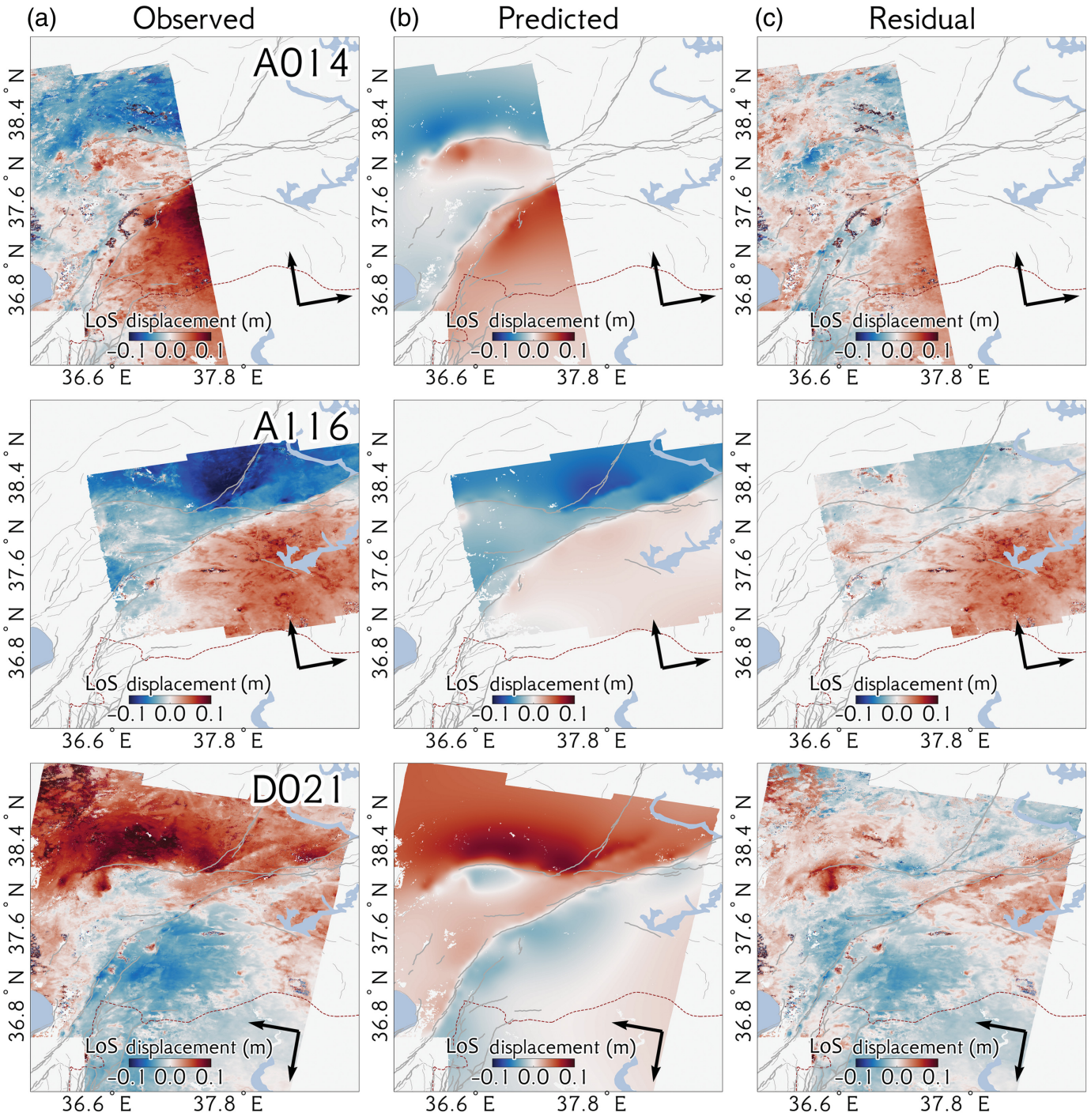


Figure 3. (a) Interferometric Synthetic Aperture Radar (InSAR) observations, (b) predictions of the mean model for the joint postseismic slip, and (c) the corresponding residuals. LoS, line of sight.

the east-northeast-trending plane that enters the NFC from the northeast. We choose to extend our model of the northern branch according to this plane. We do not include the geologically mapped fault that connects both branches, as there are no clear signs of it having been ruptured during the two events. Based on the location of the aftershocks, we also include secondary faults for both branches.

The spatial distribution of the aftershocks suggests that the northern branch is dipping significantly to the north, whereas the southern branch is well approximated by a vertical fault plane (Barbot *et al.*, 2023). We choose 70° and 90° uniform dip angles, respectively, to represent these two structures and the corresponding secondary faults, while also accounting for the possible error introduced by incorrectly chosen values (Ragon *et al.*, 2018). We assume a standard deviation of 10° for the uncertainty in the dip angle.

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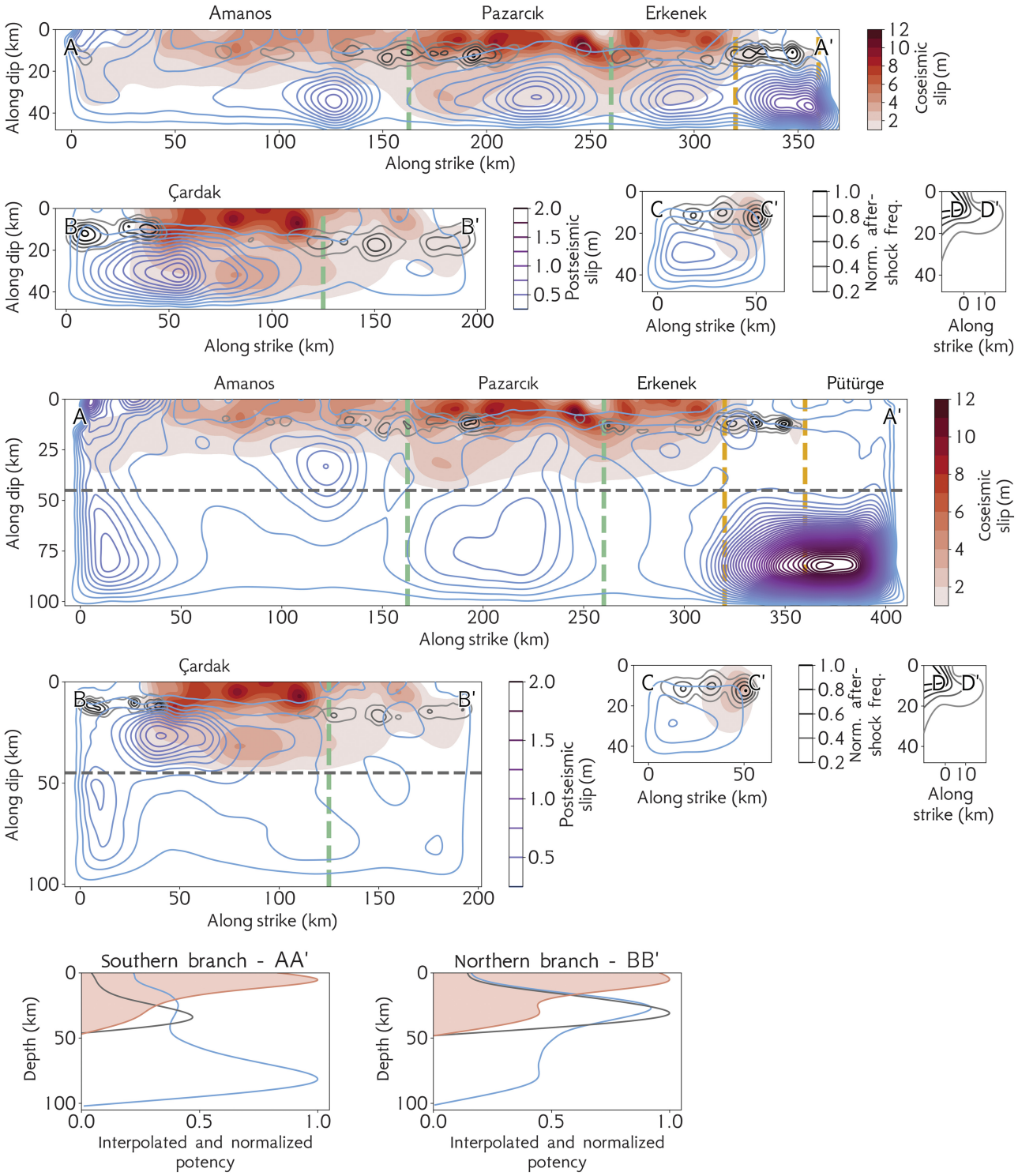
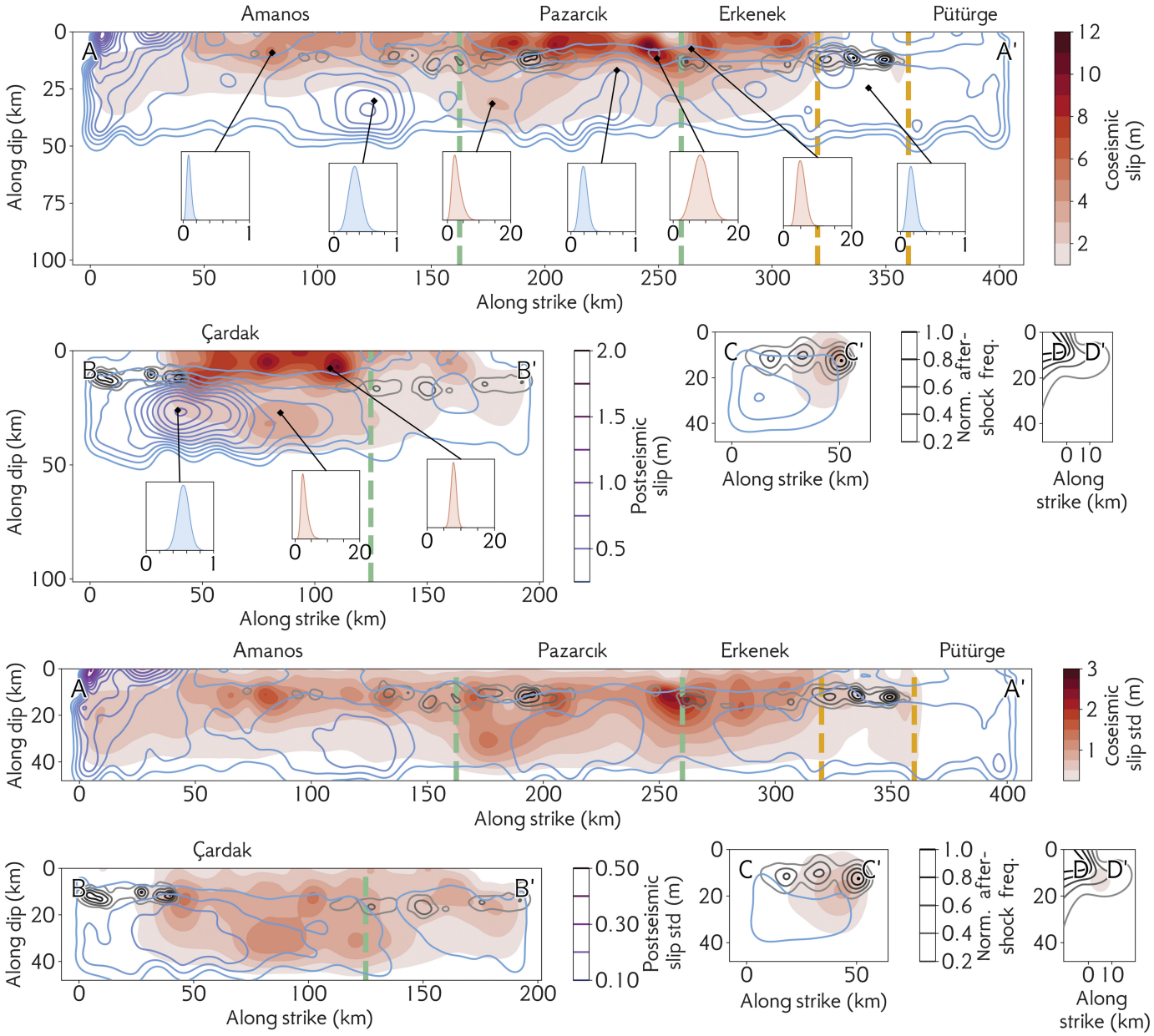


Figure 4. Posterior distribution mean of co- and postseismic slip for minimum and maximum depth fault planes. Coseismic slip on all of the models is represented with filled red contours of 1 m, starting at 1 m. Postseismic slip is displayed with blue contours of 5 cm, starting at 5 cm. Grayscale contours of 0.2, starting at 0.2, show the normalized aftershock density, taken from a relocated catalog (Lomax, 2023). The horizontal dashed gray line represents the cutoff depth that, in our opinion, separates the true afterslip from the model artifacts. Vertical dashed green lines represent boundaries between different fault segments. These boundaries are marked with diamonds in Figure 2. Vertical dashed golden lines show the approximate boundaries of the seismic gap, mentioned in Barbot *et al.* (2023), where the eastern boundary is chosen based on the aftershock and afterslip distribution of the 2020 Pütürge earthquake (Cakir *et al.*, 2023). These boundaries are also marked with diamonds in Figure 2. Potency is also plotted against depth: red indicating coseismic slip, gray and blue showing postseismic slip for the minimum- and maximum-depth fault plane models, respectively.

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We subdivide our faults into a set of triangles, where slip is described using linear shape functions centered at the vertices (nodes) of each triangle and decreasing to zero at its opposing sides (Jiang and Simons, 2016). The size of the triangles increases nonlinearly with depth to account for the rapidly decaying sensitivity of geodetic data to details of deeper subfaults. We impose a no-slip constraint on the boundary nodes, except at the surface. Our model for coseismic slip inversion has the deepest nonzero nodes located at ~35 km depth. This depth proves to be sufficient because most of the coseismic slip and aftershock activity is confined to much shallower depths; however, the same subfault geometry used for postseismic slip inversion leads to a high slip concentration along the bottom margin (Fig. 4). To investigate this behavior further, we explore two additional models. One extends the fault to ~60 km depth (Fig. S11), and another to ~85 km depth (Fig. 4). Relative to the

Figure 5. Posterior distribution mean and standard deviation of co- and postseismic slip for the maximum-depth fault plane, with everything that we do not consider to be real afterslip not shown. This is the model used to generate the predictions shown in Figures 2 and 3. For representative nodes, posterior distributions of both co- and postseismic slip are also displayed. Coseismic slip is represented with filled-red contours of 1 m, starting at 1 m. Coseismic slip uncertainty is shown with filled-red contours of 0.25 m, starting at 0.25 m. Postseismic slip is displayed with blue contours of 5 cm, starting at 5 cm. Postseismic slip uncertainty is represented with blue contours of 2.5 cm, starting at 2.5 cm. Grayscale contours of 0.2, starting at 0.2, show the normalized aftershock density, taken from a relocated catalog (Lomax, 2023). Vertical dashed green and golden lines are the same as in Figure 4.

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coseismic model, the postseismic fault model also extends further to the east, up to Lake Hazar, to investigate how far afterslip may propagate in that direction.

To clarify, Figure 4 presents slip models for the shallowest and deepest fault geometries considered (the intermediate model is provided in the supplemental material). By comparing these end-member models, we identify the shallow-slip features that remain consistent across the range of fault geometries and therefore interpret only this robust component as physically meaningful. Figure 5 presents the resulting interpreted shallow-slip model together with its associated uncertainty estimates. The corresponding model predictions are shown in Figures 2 and 3. We believe that extending the fault model to greater depths is beneficial because the long-wavelength component of the geodetic signal is present regardless of the chosen model geometry. If the model is restricted to shallow depths, the inversion will still attempt to fit these long-wavelength signals, potentially introducing spurious shallow slip. By allowing slip at greater depths, we can distinguish stable shallow-slip features from deep-slip patterns that are sensitive to the assumed fault geometry.

Elastic model

We use Green's functions for a 1D layered elastic model (Heimann *et al.*, 2017; Table S2). The number of layers in the model, their depths, as well as the corresponding seismic velocities, are based on profiles 1 and 2 of a local earthquake tomography study by Ozer *et al.* (2019), which cross the region being modeled. We choose the layer densities based on gravity modeling results (Mahatsente *et al.*, 2018). Despite these constraints, there is a high level of uncertainty in the assumed elastic model. Therefore, we adopt a 10% standard deviation in the shear modulus of each layer (Table S2). Because the shear modulus is parameterized as a log-normal variable, these uncertainties are transformed into the corresponding log-space variance used in the calculations. The Poisson's ratio is kept fixed within each layer. We follow the methodology presented in Duputel *et al.* (2014; Fig. S12).

Bayesian Inference Framework

We explore the full model space of co- and postseismic slip distributions compatible with geodetic observations. We estimate the following probability density function:

$$\rho(\mathbf{m}|\mathbf{d}) \propto \rho(\mathbf{m}) \times exp[-\chi(\mathbf{m})],$$

in which $\mathbf{m}$ is a vector of model parameters and $\rho(\mathbf{m})$ is the prior information on this sample. $\chi(\mathbf{m})$ is the misfit function that quantifies the differences between observations and predictions within the uncertainties described by the covariance matrix $\mathbf{C}_\chi = \mathbf{C_d} + \mathbf{C_p}$, in which $\mathbf{C_d}$ represents observational errors and $\mathbf{C_p}$ epistemic uncertainties introduced by approximations of the forward model (Tarantola, 2005; Minson *et al.*, 2013; Duputel *et al.*, 2014; Ragon *et al.*, 2018):

$$\chi(\mathbf{m}) = \frac{1}{2}[\mathbf{d}_{\text{obs}} - \mathbf{G}(\mathbf{m})]^{\mathrm{T}} \times \mathbf{C}_\chi^{-1} \times [\mathbf{d}_{\text{obs}} - \mathbf{G}(\mathbf{m})].$$

For a static slip inversion, the forward problem is linear and $\mathbf{G}(\mathbf{m})$ equals $\mathbf{Gm}$. Vector $\mathbf{m}$ is composed of the slip values along the strike and along the dip for every linear shape function, as well as parameters that describe a constant shift for each of the satellite data sets. We do not solve for linear or quadratic ramps for the coseismic inversion because we did not observe significant differences in the resulting models when doing so. We include the possibility of erroneous ramps in the postseismic inversion because the SNR of the data sets is smaller than for the coseismic data and much deeper subfaults are used. Neglecting these ramps can drive spurious values for deep slip.

We set uniform priors for the strike-slip components for both the co- and postseismic inversions. For coseismic inversion, slip is limited to the [0 m, 30 m] range, whereas in the case of postseismic inversion, slip is limited to the [0 m, 3 m] range. In both cases, we check that the mean values of the model parameters are within the given limits, so that no errors are introduced. Priors for dip-slip components are normal distributions with standard deviations of 1 and 0.1 m for co- and postseismic inversions, respectively.

One could sample the slip distribution using a sequential Monte Carlo (SMC) sampler, such as the one implemented in the AlTar2 package, modified from the original formulation developed by Minson *et al.* (2013). AlTar2 combines the Metropolis algorithm with a tempering process to effectively sample the model space through multiple steps. For this method, $\mathbf{C}_\chi$ is usually updated every step, based on the mean model of the Bayesian ensemble. However, we found that when $\mathbf{C}_\chi$ is too large, our postseismic slip inversion based on the SMC sampler can become unstable, which then leads to unphysical updates to $\mathbf{C_p}$ and drives an unwanted feedback loop, causing the final model to systematically overfit the observables.

To address this problem, we use an alternative method to calculate $\mathbf{C_p}$, which is not dependent on the SMC updates and thus could also be used with a more efficient sampler. We start with the $\mathbf{C_d}$-only inversion and use the resulting model to calculate $\mathbf{C_p}$, which in this case has a much more reasonable amplitude, constrained by the data observations. We then run a series of inversions, updating $\mathbf{C_p}$ every time with the model that results from averaging models of all the previous steps. This process results in a clearly converging sequence (Figs. S13 and S14). We stop at 50 iterations because by this step any model changes are insignificant (Figs. S13 and S14). The instability does not occur in this case because the sampler does not necessarily need to go through the tempering process. For these inversions, we use a Hamiltonian Monte Carlo (HMC) method in its no U-turn sampler (NUTS) formulation (Hoffman and Gelman, 2014; Betancourt, 2017), as implemented in the PyMC package Abril-Pla *et al.* (2023).

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Gradient-based samplers are especially well suited for linear problems. Currently, the most efficient general-purpose gradient-based samplers are based on HMC. These can be developed by defining a dynamical system where the likelihood of the distribution we want to study is associated with the potential energy of the system, which is then simulated using Hamiltonian dynamics. That is, HMC borrows an idea from physics to suppress the local random walk behavior in the Metropolis algorithm, allowing it to move much more rapidly through the target distribution (Gelman *et al.*, 1995). The original version of HMC has several algorithmic parameters that need to be set and fine-tuned, that is, mass matrix (preconditioner), step size, and number of steps that are used for simulating a particle motion. However, NUTS automatically finds the optimal number of steps, whereas packages such as PyMC (Abril-Pla *et al.*, 2023) or Stan Development Team (2016) also adaptively choose the other two hyperparameters. The method, while well established, has only recently started to be used in geophysics (Fichtner and Simutė, 2018; Fichtner *et al.*, 2019; Yamada *et al.*, 2022). We also compare the resulting coseismic model distribution to one obtained through the Bayesian approach as implemented in the AlTar2 package for validation purposes (Fig. S15). The distributions are practically equivalent; however, HMC requires significantly less computational time. For the HMC sampler we run four chains in parallel for 2500 steps as a burn-in period and then for 2500 more steps that are eventually used as an ensemble of 10,000 models for the later analysis. We use an ensemble of 100,000 models for the SMC sampler in the AlTar2 package.

Results

Model postprocessing

In contrast to regularized least-squares inversions, the Bayesian framework provides an ensemble of models. From the posterior distribution of every parameter, we can compute the mean, median, or maximum a posteriori probability estimate. We consider the mean for all the models presented in the main text; however, the median (Figs. S16 and S17) and maximum a posteriori estimate models (Figs. S18 and S19) are consistent with the former, as shown in the supplemental material. One can also smooth the posterior model ensemble, which should not be confused with a priori regularization. In our case, we found both the co- and postseismic mean models to be quite rough (Figs. S20 and S21) and the corresponding posterior distributions resembling Dirichlet distributions, indicating that our data do not have enough resolution to distinguish between slip at neighboring locations (Duputel *et al.*, 2014; Jiang and Simons, 2016). One could use a spatial averaging operator that averages the value of slip at each mesh node with all its nearest neighbors, while weighting those values by the area covered by the corresponding linear shape functions (Jiang and Simons, 2016) or giving them a uniform weight. We used the latter approach, but with less posterior smoothing for the coseismic model. That is, we divided the weights of the neighboring nodes by the overall number of neighbors. After posterior spatial averaging, we use the mean for later analysis and presentation. We emphasize that the possibility of applying smoothing a posteriori could be considered a major advantage of the Bayesian approach. We do not claim to have found an optimized smoothing approach for our case. Indeed, the smoothing operators are arbitrary and slightly different for the co- and postseismic cases. Our motivation for a particular spatial averaging operator has been based on the desire to have posterior distributions for individual parameters that are more Gaussian-like and not Dirichlet-like, while still displaying some finer details (Duputel *et al.*, 2014).

Coseismic slip distribution

The Amanos and Pazarcık faults on the southern branch and the Çardak fault on the northern branch experienced the most slip throughout the seismic events (seismic moments of 4.1×10^{20}, 4.4×10^{20}, and 4.9×10^{20} N · m, respectively), whereas the Erkenek fault has contributed approximately half as much as each of the other segments to the overall seismic moment of the event (2.4×10^{20} N · m). Minimal slip occurs on the continuation of the Çardak fault beyond the NFC. These models correspond to magnitudes M_w 8.0 and 7.9 for the earthquakes of the northern and southern branches.

We observe clear differences in slip patterns across multiple segments. For the southern branch (**AA′** in Figs. 4 and 5), the slip distribution is as follows. Slip on the Amanos fault is consistently shallow along the whole length, barely reaching a depth of 20 km, with an average slip in the 2–5 m range. The Pazarcık fault hosts the highest coseismic slip on this branch, exceeding 10 m. Deep slip is also observed at a depth of 35 km at the boundaries between different segments. Further to the east, a transition to the Erkenek fault occurs with intermediate values of slip of 4–7 m.

For the northern branch (**BB′** in Figs. 4 and 5), the difference of slip distribution between the two segments is also clear. The Çardak fault has consistently high slip values, exceeding 7 m, releasing more energy than the Pazarcık fault on a shorter segment. Slip on the fault across the NFC is, on the contrary, very shallow, mostly constrained to the upper 10 km, and reaches values of only a few meters. We also observe deep coseismic slip along the Çardak fault. This deep slip is not evident in other existing models because their fault planes are limited to the depth of ∼20 km.

Postseismic slip distribution

It is apparent from the postseismic slip distribution that a significant portion of the data sets cannot be explained within the limitations of our elastic modeling (Figs. 2 and 3). The inverted slip separates itself into two parts: one that is stable and localized proximal to the coseismic slip, and another component that is unstable and shifts down as we increase the maximum depth to which slip is allowed (Fig. 4).

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There are consistent components of deep slip along strike. On the southern branch, for the deepest fault model we observe a postseismic slip of up to 40 cm in the bottom-left edge of the fault plane and up to 25 cm below the Pazarcık fault, as well as a significant slip of up to 190 cm in the bottom-right edge. On the northern branch, there is slip of up to 25 cm observed in the bottom-left corner and up to 10 cm in the bottom-right corner (Fig. 4).

On the southern branch, we find significant shallow postseismic slip of up to 75 cm close to the western margin of the Amanos fault (Fig. 4). The model also shows some afterslip below the coseismic rupture zone, which is prominent along the three faults (Fig. 4). The amplitude of this shallow afterslip is modest, reaching values of only 35, 20, and 20 cm along the Amanos, Pazarcık, and Erkenek faults, respectively. No significant afterslip is observed for the seismic gap except for the small patch of up to 25 cm slip at the edge of the coseismic rupture (Fig. 4). We infer a much higher afterslip of up to 55 cm for the northern branch.

Uncertainties

The analysis of coseismic uncertainties (Fig. 5) suggests that geodetic data sets are rapidly losing their ability to accurately constrain slip at depths greater than 10 km. It follows that the highest coseismic uncertainties belong not to the shallowest regions with the highest slip, but to the deeper portions of the model. A portion of coseismic slip in our model is smeared from above, and the actual slip versus depth profile might be sharper, with most actual slip occurring above 10 km. For the deeper coseismic slip areas, the relative uncertainties are fairly high. On the southern branch, slip values reach 3 m, while the uncertainties are 1.25 m. On the northern branch, the maximum deep coseismic slip is 4 m, whereas the corresponding uncertainties are above 1 m. We think that these values suggest the credibility of the existence of deep coseismic slip (or rapid afterslip, captured by the satellite data sets), whereas the amplitudes of slip might be biased. The postseismic slip for the smoothed model is also constrained fairly well. One could see (Fig. 5) that the uncertainties for the southern branch are on the order of 5–7.5 cm for the regions with notable afterslip, whereas the corresponding mean postseismic slip is on the order of 20–35 cm. For the northern branch, the values of postseismic slip reach 55 cm, whereas the corresponding uncertainties are 7.5 cm.

Discussion

Our coseismic slip distribution is consistent with previous models, especially those that had used a similar combination of geodetic data sets, for example, He *et al.* (2023). Using our Bayesian approach, we find that the junctions of different fault segments, marked with green-dashed lines in Figures 4 and 5, have lower slip values relative to their surroundings. At the same time, the slip within each fault segment is relatively smooth and continuous. The two branches could be divided into multiple segments based solely on the slip distribution, which suggests that the characteristic of rupture is strongly dependent on the pre-existing tectonic structures of the region.

Our estimates of coseismic moment magnitude are larger than those that have been presented before. Presumably because we allow slip to extend below 20 km, which is not the case for previously published models. The deep coseismic slip, as well as the possible early afterslip that cannot be separated in the satellite data sets, may contribute to the apparent increase in the moment magnitude of the events. The deep-slip regions on the southern branch, located at the boundaries between the different fault segments (the westernmost region is also close to the junction with the secondary fault), might be an artifact of our model simplifications. That is, relatively simple fault planes might not be able to represent fault junctions accurately. However, the anticorrelation of those areas of deep coseismic slip with areas of the aforementioned postseismic shallow afterslip is evident, which suggests that this coseismic slip may be real. The deep coseismic slip on the northern branch is not close to any junction and is not localized. Thus, we assume that it is likely real.

With respect to the distribution of fault slip on the northern branch, one could attribute the difference between the high slip values at depth on the Çardak fault and the shallow low slip values on its continuation to the recent formation of the fault across the NFC. Observations made by Perrin *et al.* (2016) suggest that the largest coseismic slip for continental earthquakes is systematically produced on fault sections embedded in the narrowest but most compliant damage zones. Those sections are also the most structurally mature, where the maturity of the fault qualitatively indicates the degree of advancement in the evolution of its structural properties such as the initiation age, net slip, length, and slip rate (Perrin *et al.*, 2016).

Although we cannot rule out the possibility of contamination of the whole solution due to off-fault processes, the apparent deep postseismic slip most likely represents viscoelastic volumetric processes that are mismodeled in our purely elastic afterslip model. Similar observations have been made in the studies of the 2002 Denali earthquake in Alaska (Freed *et al.*, 2006; Biggs *et al.*, 2009). Thus, we only give physical interpretation to the stable part of the solution, which we assume reflects true afterslip (Fig. 5). It should be noted that, in our view, allowing the fault plane to extend to greater depths is the more appropriate approach in the absence of a full viscoelastic model. The long-wavelength signal responsible for the artificial deep slip remains present in the data regardless of the adopted fault geometry, and the inversion will attempt to fit it. Therefore, restricting the fault plane to shallower depths would increase these biases, as the model would be forced to accommodate a fundamentally long-wavelength signal within a limited depth range.

The high-amplitude afterslip on the western margin of the Amanos fault is possibly related to the aftershock of 20

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February 2023 M_w 6.4 (Barbot *et al.*, 2023), which we do not model explicitly. The general afterslip pattern along the southern branch agrees with previous studies of postseismic slip of major earthquakes (Ryder *et al.*, 2007; Biggs *et al.*, 2009; Hearn *et al.*, 2009; Wang and Fialko, 2014; Hong and Liu, 2021; Wang *et al.*, 2021), areas of high co- and postseismic slip do not overlap, and postseismic slip is observed in our case up to the depths of 45 km, further down-dip than the coseismic slip. The postseismic slip distribution also matches the results of Guo *et al.* (2025), whose kinematic afterslip model has been produced using similar data.

The seismic gap described in Barbot *et al.* (2023) is outlined by golden vertical lines in Figures 4 and 5. The shallowest model, the same as the one used for coseismic inversion, incorrectly merges the deep and shallow components of the slip together (Fig. 4). Considered in isolation, this model could be interpreted as indicating that afterslip extends continuously across the junction between the Pazarcık and Elazığ faults. However, comparison with models that employ progressively deeper fault geometries reveals that the shallow slip pattern remains essentially unchanged, whereas the deeper slip systematically migrates downward as additional fault depth is introduced. This sensitivity to the assumed fault geometry indicates that the deep slip is not robust and is most likely an inversion artifact arising from attempts to fit long-wavelength components of the data. In contrast, the shallow slip is stable across all tested models and is therefore interpreted as physically meaningful. Consequently, we conclude that afterslip is not reliably resolved across the gap.

In general, if one considers the postseismic potency distribution versus depth (Fig. 4), it is clear that neglecting the depth extension of the fault plane introduces a bias. Again, we consider it more appropriate to allow slip to extend deeper along the fault plane, while interpreting only the stable patterns as physically meaningful. In the models that do not extend as deeply, for the southern branch the pattern changes significantly: the shallow afterslip is no longer concentrated around 40 km depth, separated from the coseismic slip, but is shifted upward, closely overlapping and "supplementing" the coseismic slip, with a maximum afterslip reached at a depth of 30 km. For the northern branch, the bias is less significant, and the overall location of the afterslip is similar for both models. However, the extended fault-plane model has the afterslip at a slightly shallower depth. It also overlaps significantly with coseismic slip. The coseismic potency distribution as a function of depth shows local complexities at 20–40 km. We interpret them as rapid afterslip.

The distribution of the afterslip on the northern branch might indicate that a narrow fault zone along this branch extends deeper through the crust and possibly into the upper mantle. We also infer deep coseismic slip. Although it could be an artifact of modeling, we interpret this slip as potentially representing early afterslip, not covered by the InSAR time series used in the postseismic inversion but captured in the coseismic data. This interpretation is supported by the fact that the deep coseismic slip is not overlapped by the later postseismic slip. Based on the joint distribution of these slip categories, one could infer that the Çardak fault has a well-defined deep fault zone.

Heightened aftershock activity mostly either extends away from the coseismic rupture zone on its boundaries, penetrating the non-ruptured neighboring segments, or is restricted to the gaps in between areas of high slip, thus evening out the energy release. This filling of the unruptured gaps is especially well observed on the southern branch at about 195 and 265 km along the strike (Figs. 4 and 5).

Conclusion

We present a model of co- and postseismic static slip distributions in a Bayesian framework, taking into account the estimation of uncertainties, both observational and epistemic. Our coseismic slip model is consistent with previously published models. The avoidance of any prior smoothing enables a more detailed view of the interaction between coseismic slip and immediate aftershocks. Rupture, propagating across the complicating fault structures, changes its characteristics, such as the mean slip value and maximum slip depth, which illustrates the importance of preexisting fault complexities. The postseismic geodetic data cannot be well modeled with only elastic afterslip and may require consideration of viscoelastic behavior. The stable component of the afterslip in our model is mostly deep (around 30 km), surrounding the areas of coseismic rupture. The joint analysis of the coseismic slip and the afterslip on the southern branch emphasizes the lateral variability in the depth of coseismic slip, with coseismic and the early postseismic slip complementing each other. For the northern branch, a deep and narrow fault zone is suggested, which has not been yet developed, though, beyond the Çardak fault. We reaffirm the possibility of a seismic gap on the boundary between the Erkenek and Pütürge faults. Although the study presents a robust and well-constructed inversion framework for modeling co- and postseismic slip, the depth resolution of the inferred slip distributions warrants careful interpretation. The deep slip components may be influenced by epistemic uncertainties in the elastic model or artifacts related to the choice of fault geometry and data covariance used.

Data and Resources

Copernicus Sentinel-1 data have been acquired by the European Space Agency (ESA). These products are archived at the Alaska Satellite Facility (ASF; https://search.asf.alaska.edu/ and https://s1qc.asf.alaska.edu/ for data and orbit products, respectively). The Advanced Land Observation Satellite-2 (ALOS-2) data have been collected by the Japan Aerospace Exploration Agency (JAXA). These products are proprietary and cannot be released to the public. The Global Navigation Satellite Systems (GNSS) coseismic offsets could be obtained from the UNR Geodetic

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Laboratory website (https://geodesy.unr.edu/news_items/20230213/us6000jllz_final5min.txt and https://geodesy.unr.edu/news_items/20230213/us6000jlqa_final5min.txt). Daily GNSS solutions, which have been used to estimate the postseismic offsets, are available from the UNR Geodetic Laboratory website as well (https://geodesy.unr.edu/NGLStationPages/GlobalStationList). The Turkish Seismic Network Catalogue was searched using https://deprem.afad.gov.tr/event-catalog. All the other data used in this article came from published sources listed in the references. All websites were last accessed in October 2026. The supplemental material for this article includes: dates of the Synthetic Aperture Radar (SAR) pairs used for the coseismic slip inversion; description of the 1D layered elastic half-space model used in the inversion; downsampled coseismic pixel offsets and postseismic Interferometric Synthetic Aperture Radar (InSAR) observations along with the corresponding predictions and residuals; Gaussian model fits to the variogram clouds for coseismic pixel offsets and postseismic InSAR data; posterior distribution mean of smoothed co- and postseismic slip for the models with unmodified GNSS uncertainties; posterior distribution mean of smoothed coseismic slip for the model obtained using the AlTar2 package; logarithmic model fits to the observations for postseismic GNSS data; perpendicular baseline versus acquisition time plot for the tracks used in the postseismic slip inversion; posterior distribution mean of smoothed postseismic slip for the intermediate depth fault-plane model; normalized model distances for co- and postseismic models during multiple $\mathbf{C_p}$ iterations relative to the $\mathbf{C_d}$-only model; posterior distribution median and maximum a posteriori estimate of smoothed co- and postseismic slip models; posterior distribution mean of unsmoothed co- and postseismic slip models.

Declaration of Competing Interests

The authors acknowledge that there are no conflicts of interest recorded.

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